



# UTM

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**BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)**

**SECJ3553 ARTIFICIAL INTELLIGENCE - SECTION 01**

**PROJECT ASSIGNMENT 2**

**THEME: HEALTHCARE**

**( MINDCARE AI - INTELLIGENT MENTAL HEALTH SUPPORT CHATBOT )**

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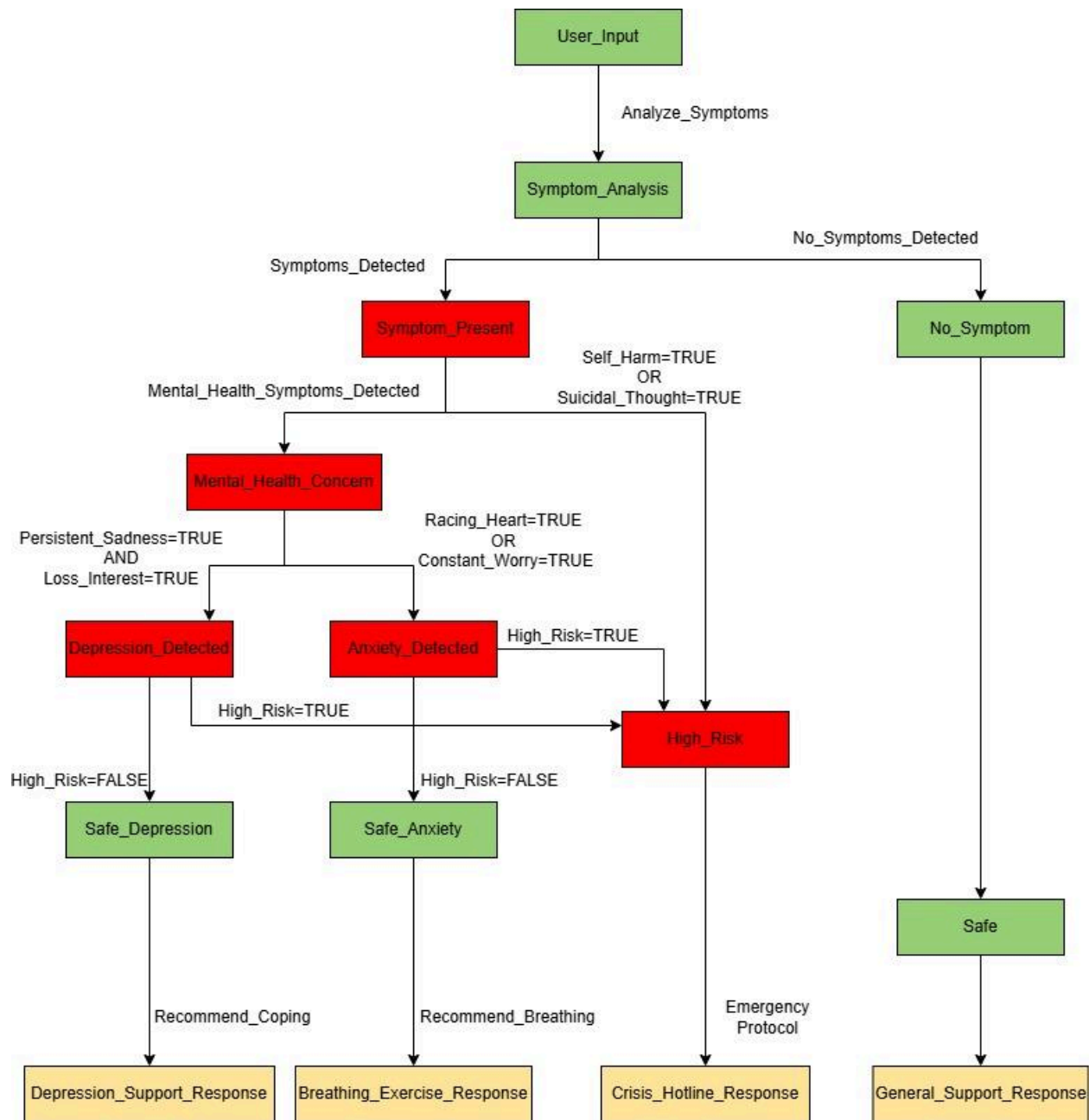
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## **Overview**

This state space search builds upon the five Knowledge Representations (KRs) established in Assignment 1 to model how the MindCare AI chatbot processes user inputs, evaluates mental health conditions, and provides appropriate responses.

## State Space Search



## States Definition

| State                 | Definition                                                                                                                                                                                                                                                                            |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User_Input            | The initial state where the user provides text input describing their emotional or mental state through the chatbot interface. This serves as the entry point for all interactions.                                                                                                   |
| Symptom_Analysis      | The state where the system processes and analyzes the user's input using Natural Language Processing (NLP) to identify keywords and patterns related to mental health symptoms.                                                                                                       |
| No_Symptom            | A safe state indicating that no mental health symptoms (persistent sadness, loss of interest, racing heart, constant worry, self-harm, or suicidal thoughts) were detected in the user's input. The user may be seeking general wellness information or expressing positive emotions. |
| Symptom_Present       | A critical detection state indicating that one or more mental health symptoms have been identified in the user's input. This triggers further classification into specific conditions.                                                                                                |
| Mental_Health_Concern | A general state indicating that mental health symptoms (depression or anxiety related) have been detected, but no immediate safety risk. This serves as a classification checkpoint before determining specific condition type.                                                       |
| Depression_Detected   | A condition state where the system has identified both persistent sadness AND loss of interest in the user's input, meeting the criteria for depression signs according to KR1.                                                                                                       |
| Anxiety_Detected      | A condition state where the system has identified either racing heart OR constant worry in the user's input, meeting the criteria for anxiety signs according to KR2.                                                                                                                 |
| High_Risk             | A critical emergency state where the system has detected mentions of self-harm OR suicidal thoughts, indicating immediate safety concerns. This is the highest priority state in the system according to KR3.                                                                         |
| Safe_Depression       | A manageable state where depression symptoms are present but no high-risk indicators are detected. The user can be provided with coping strategies and supportive resources.                                                                                                          |
| Safe_Anxiety          | A manageable state where anxiety symptoms are present but no high-risk indicators are detected. The user can be provided with calming techniques and breathing exercises.                                                                                                             |
| Safe                  | A positive state for users with no detected symptoms who will receive general mental wellness support, mood tracking suggestions, and preventive mental health education.                                                                                                             |
| Depression_Support_Re | A response state where the system provides empathetic support, coping                                                                                                                                                                                                                 |

|                             |                                                                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| sponse                      | strategies, and professional help recommendations specifically tailored for depression symptoms.                                                                               |
| Breathing_Exercise_Response | A response state where the system guides the user through calming breathing exercises and anxiety management techniques according to KR4.                                      |
| Crisis_Hotline_Response     | An emergency response state where the system immediately provides crisis hotline contact information, emergency resources, and urgent support messages according to KR5.       |
| General_Support_Response    | A supportive response state where the system provides general mental wellness tips, mood tracking suggestions, and positive reinforcement for users with no detected symptoms. |

## Actions Definition

| Action                                         | Definition                                                                                                                                                                                                                                                                     |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analyze_Symptoms                               | The NLP-based process that examines user input to extract and identify mental health symptoms such as persistent sadness, loss of interest, racing heart, constant worry, self-harm mentions, or suicidal thoughts. This action uses sentiment analysis and keyword detection. |
| Symptoms_Detected                              | The action triggers when the system identifies one or more mental health symptoms in the user's input, moving the state from Symptom_Analysis to Symptom_Present for further classification.                                                                                   |
| No_Symptoms_Detected                           | The action triggered when the system finds no mental health symptoms in the user's input, routing the user to the safe path for general wellness support.                                                                                                                      |
| Persistent_Sadness=TRUE AND Loss_Interest=TRUE | The logical evaluation action that checks if both persistent sadness AND loss of interest symptoms are present simultaneously, as defined in KR1 for depression identification.                                                                                                |
| Racing_Heart=TRUE OR Constant_Worry=TRUE       | The logical evaluation action that checks if either racing heart OR constant worry symptoms are present, as defined in KR2 for anxiety identification.                                                                                                                         |
| Self_Harm=TRUE OR Suicidal_Thought=TRUE        | The critical safety assessment action that checks if any mentions of self-harm OR suicidal thoughts are present, as defined in KR3 for high-risk classification. This action takes the highest priority.                                                                       |
| High_Risk=FALSE                                | The risk classification action that confirms the absence of self-harm or suicidal thoughts, allowing the system to provide non-emergency support for depression or anxiety.                                                                                                    |
| High_Risk=TRUE                                 | The risk classification action that confirms the presence of self-harm or suicidal thoughts, immediately routing the user to emergency crisis                                                                                                                                  |

|                       |                                                                                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | intervention regardless of other conditions.                                                                                                                                                          |
| Provide_Wellness_Tips | The action that generates general mental health education, mood tracking recommendations, and wellness guidance for users with no detected symptoms.                                                  |
| Recommend_Coping      | The action provides depression-specific coping strategies, professional help suggestions, and supportive resources for users with manageable depression symptoms (KR4 applied to depression context). |
| Recommend_Breathing   | The action that guides users through breathing exercises and calming techniques specifically for anxiety management, as defined in KR4.                                                               |
| Emergency_Protocol    | The critical action that immediately provides crisis hotline information, emergency contact numbers, and urgent support resources as defined in KR5. This action overrides all other recommendations. |

## Problem Formulation

We formulate the MindCare AI reasoning process as a state space search problem defined by four key components:

### i) Initial state

Description: The search begins when the system receives a raw text input from the user describing their mental state. At this stage, no processing or analysis has occurred

- State: User\_Input

### ii) Action

Description: The available actions are the logical transitions and condition checks derived from the Knowledge Representation rules. The successor function returns the set of legal actions for any given state:

- Processing Actions: Analyze\_Symptoms, Symptoms\_Detected, No\_Symptoms\_Detected
- Logic/Diagnostic Actions: Mental\_Health\_Symptoms\_Detected, Check\_Depression, Check\_Anxiety, Check\_Risk, Check\_Safety
- Recommended Actions: Recommend\_Coping, Recommend\_Breathing, Emergency\_Protocol

### iii) Goal

Description: The search terminates when the agent reaches a valid Response State. The goal is to reach one of the following states:

- Depression\_Support\_Response
- Breathing\_Exercise\_Response
- Crisis\_Hotline\_Response
- General\_Support\_Response

### iv) Path cost

Description: Each transition represents one unit of logical inference or processing. The objective is to find a path from the Initial State to a Goal State with the minimum number of steps necessary to provide an accurate response.

- Step cost = 1
- Total path cost = Total number of edges traversed from the start



## **Solution**

Below is a solution trace for a specific scenario: "Anxiety Detected (Safe)".

Scenario: The user mentions "racing heart" or "worry" but does not indicate any self-harm.

Solution sequence:

Start State: User\_Input

Action 1: Analyze\_Symptoms → Symptom\_Analysis (State)

Action 2: Symptoms\_Detected → Symptom\_Present (State)

Action 3: Mental\_Health\_Symptoms\_Detected → Mental\_Health\_Concern (State)

Action 4: Check\_Anxiety → Anxiety\_Detected (State)

Action 5: Check\_Safety → Safe\_Anxiety (State)

Action 6: Recommend\_Breathing → Breathing\_Exercise\_Response (Goal State)

Total path costs : 6 steps